Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> Power (P) can be calculated using the formula: P = IV, where P is power in watts, I is current in amperes, and V is voltage in volts. We also know that Power can also be calculated using the formula: P = IÂ²R, where P is power in watts, I is current in amperes, and R is resistance in ohms. We can use the power formula to solve for current (I), and then use the current to solve for resistance (R). We know the power is 40 watts and the voltage is 12 volts. Let's solve for the current first: 40 = IÂ² ÃĹ 12. Dividing both sides by 12, IÂ² = 3.33. Taking the square root gives us I = 1.83 A. Now we can find the resistance: 40 = 1.83Â² ÃĹ R. Solving for R, we find R = 10/1.83. Using a calculator, R âīĪ 5.45 ohms. </think>

<answer> H. 2.2 amps, 5.45 ohms </answer>